# Jayanth Vegesna

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## Summary

First-year postgraduate Biomedical Engineering student with interdisciplinary research experience spanning computational biophysics, molecular parasitology, and infectious disease biology. Experienced in molecular dynamics simulations, molecular cloning, and bioinformatics workflows, with research experience at IIT Mandi, IBAB Bengaluru, and NHRI Taiwan. Interested in research in tissue engineering and 3D bioprinting.

## Education

**Master of Technology Biomedical Science and Engineering** *Jul 2026 - May 2028*
*Indian Institute of Technology Guwahati* Guwahati, Assam
Specialisation: Regenerative Medicine, Stem Cells & Therapeutics

**Bachelor of Technology Biotechnology** *2022 - 2026*
*Lovely Professional University* Phagwara, Punjab
CGPA: **8.98 / 10**

## Research Experience

**Research Intern (Molecular Parasitology Lab)** *Aug 2025 - Present*
*Institute of Bioinformatics and Applied Biotechnology (IBAB)* Bengaluru, Karnataka

- Contributing to a study on R-loops in *Plasmodium falciparum* by cloning and purifying PfRNaseH enzymes via infusion cloning and recombinant expression in *E. coli*.
- *In silico* characterisation of putative PfRNaseH2A-C subunits including solubility, phase-separation, and intrinsically disordered region (IDR) prediction.
- Utilised CLANS and EFI-EST to construct sequence similarity networks studying the divergence of PfRNaseH2-C across the eukaryotic domain.

**Summer Research Intern (Computational Biophysics Lab)** *Jun 2025 - Aug 2025*
*Indian Institute of Technology (IIT) Mandi* Kamand, Himachal Pradesh

- Executed and analysed 15+ coarse-grained (MARTINI) and all-atom MD simulations (GROMACS) ranging from 500 ns to 3 $\mu$s on IIT Mandi's HPC cluster.
- Simulated protein-membrane, protein-RNA, membrane-only, and protein-ligand systems; performed PCA analyses using VMD and ChimeraX.
- Contributed to three PhD projects: SARS-CoV-2 polymerase RNA replication, extracellular vesicle protein stability in cancer metastasis, and hepatoprotective phytocompound-protein interactions.

**Research Intern (Division of Infectious Disease)** *Jun 2024 - Aug 2024*
*National Health Research Institutes (NHRI)* Miaoli County, Taiwan

- Worked in a BSL-2 facility comparing infection rates across three Dengue virus (DENV) genotypes via plaque assays, virus titration, and animal cell culture.
- Performed mouse procedures (necropsy, cardiopuncture, submandibular blood collection), genotyping PCR, Oxford Nanopore flow-cell preparation, IFA, and Western blotting.
- Assisted with viral RNA purification, cDNA synthesis, and flow cytometry of adenovirus-infected cells.

## Projects

**Plasmodium falciparum Multi-Omics Portal** **[jayanth-vegesna.github.io/Plasmodium]** *May 2026*

- Developed a web platform integrating data from PlasmoDB, UniProt, STRING, NCBI, and AlphaFold DB to create interactive visualisations of the proteome, genome, transcriptome, interactome, spatial localisation, and gene ontology of *Plasmodium falciparum*.

**Automated Molecular Docking Pipeline** *Oct 2024*

- Developed a reproducible Python-based molecular docking workflow integrating AutoDock Vina, MDAnalysis, and ProLIF for automated protein-ligand interaction analysis and visualization.

**Phytocompound Repurposing for Gastric Cancer** *Apr 2024*

- Screened 30+ phytocompounds against gastric cancer receptors (SwissADME); performed molecular docking for binding-affinity ranking and ran MD simulations of the highest ranked ligands computing RMSD, RMSF, SASA, and radius of gyration.

## Skills

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| **Programming** | Python, Bash, Linux, MATLAB, C/C++ |
| **Computational** | GROMACS, AutoDock Vina, MDAnalysis, Biopython, VMD, ChimeraX, CLANS, Cytoscape |
| **Wet Lab** | Molecular cloning, Protein purification, PCR, Western blotting, Bacterial cell culture, Mammalian cell culture, Flow cytometry, IFA |

## Publications

- Cherian J, Kunigiri S, **Vegesna J**, Yadav PK, Kumar V. "Geroprotective effects of bioactive compounds against hydroquinone toxicity in *Drosophila melanogaster*: an in vivo and in silico insight." *Biogerontology*, 2026. doi:10.1007/s10522-026-10452-x

## Conferences

- Attendee - 2024 Biomedical Research Symposium, NHRI Taiwan *Aug 2024*
- Poster Presenter - International Conference on Microbial Bioprospecting Towards SDGs *Nov 2023*
- Poster Presenter - International Conference on Bioengineering and Biosciences (ICBB-2022) *Nov 2022*

## Achievements

- **3rd Prize** - Manipal Biopharma Hackathon 2025, Manipal-GOK Bioincubator *Mar 2025*
- **1st Prize** - COGNITA'25 Pharma Insight Poster Presentation, LPU *Feb 2025*
- **2nd Prize** - SPARK Innovation Challenge, Regional Centre of Biotechnology *Jan 2025*
- **1st Prize** - Biofluence Bio-Hackathon, Bversity *Aug 2024*
- **1st Prize** - Poster Presentation, ICBB-2022 *Nov 2022*

## Certifications

- Research Minisymposium: "Advancing Genome Editing" - *IBAB, Bengaluru* *Feb 2026*
- Workshop on Micro-fabrication and Biosensors: Advances in Diagnostics - *IIT Jammu* *Jan 2025*
- Pandemic and Epidemic-Prone Diseases - *WHO* *Jul 2024*
- Quantitative Methods for Biology (MATLAB) - *HarvardX* *Oct 2023*
- Bioelectrochemistry - *IIT Kanpur* *Mar 2023*

## Leadership & Extracurriculars

- **President, Readers Arcadia** - Directed 4 university-wide literary and oratory events attracting 200+ participants.
- **Member, Bioaffinities** - Co-organised 2 technical workshops training 80+ peers in life science skills.
- **BIOCLUES** - Active member of a national bioinformatics professional network for research sharing and scientific discourse.